

CAPE TOWN INTL, South Africa

WMO: 688160

Lat: 33.9631S

Lon: 18.6022E

Elev: 138

StdP: 14.62

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.3	41.6	33.5	28.2	56.5	35.8	30.9	53.5	30.0	57.4	27.1	57.4	3.5	50	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	17.6	89.4	67.6	85.8	66.7	82.6	65.7	70.2	82.4	69.1	80.5	68.0	78.5	12.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	98.1	73.2	65.4	94.3	72.4	64.4	90.9	71.7	34.3	82.8	33.2	80.7	32.3	79.1	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.7	26.9	24.3	34.6	97.5	1.8	2.8	33.3	99.5	32.2	101.2	31.2	102.8	29.9	104.8
			33.7	73.2	1.9	1.9	32.4	74.5	31.3	75.6	30.2	76.7	28.9	78.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	71.1	71.4	68.7	64.3	59.8	55.8	54.7	55.5	58.1	62.3	65.4	69.2	(6)
	DBStd	7.24	3.89	3.91	4.26	4.69	4.07	3.71	3.89	3.73	4.21	4.88	4.28	3.96	(7)
	HDD50	9	0	0	0	0	0	2	5	2	0	0	0	0	(8)
	HDD65	1526	3	2	11	68	170	277	319	296	212	113	46	9	(9)
	CDD50	4746	654	599	579	429	304	176	152	172	244	381	461	595	(10)
	CDD65	786	192	181	125	47	8	1	1	1	6	29	57	138	(11)
	CDH74	4827	1071	1021	717	355	102	31	31	37	96	271	345	750	(12)
	CDH80	1327	302	304	214	110	23	3	2	5	32	69	83	180	(13)
(14) Wind	WSAvg	11.5	14.7	14.0	12.1	10.1	8.8	9.1	9.1	9.8	10.7	11.8	13.5	14.1	(14)
(15) Precipitation	PrecAvg	20.0	0.4	0.5	0.7	1.6	2.6	3.7	3.2	3.0	1.7	1.2	0.8	0.6	(15)
	PrecMax	29.5	2.3	2.1	2.8	7.0	5.7	9.1	8.2	8.5	5.4	4.3	3.4	3.5	(16)
	PrecMin	11.7	0.0	0.0	0.0	0.2	0.2	0.9	0.0	0.6	0.3	0.0	0.0	0.0	(17)
	PrecStd	4.4	0.4	0.6	0.6	1.2	1.4	1.8	1.7	1.5	1.0	1.0	0.8	0.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.6	92.3	93.2	89.7	83.0	78.7	78.2	79.1	84.3	86.8	86.8	89.6	(19)
		MCWB	68.3	68.8	67.7	65.1	61.2	58.0	57.9	58.9	61.3	63.7	65.2	68.5	(20)
	2%	DB	86.9	87.8	85.3	82.6	75.8	71.4	71.8	71.5	75.1	80.7	82.1	84.5	(21)
		MCWB	68.3	68.5	66.6	64.2	60.0	56.2	56.3	56.7	59.7	62.6	64.1	67.0	(22)
	5%	DB	82.8	83.9	80.8	77.3	71.4	66.3	66.4	66.3	69.9	76.5	77.4	80.9	(23)
		MCWB	67.3	67.3	65.6	62.5	59.3	56.0	54.9	55.9	58.6	61.7	62.9	66.1	(24)
	10%	DB	79.2	80.2	77.3	73.4	67.8	62.9	62.8	62.9	66.1	71.8	73.8	78.0	(25)
		MCWB	66.3	66.9	65.1	62.0	59.0	55.6	55.0	55.7	57.2	60.5	62.0	65.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.4	72.2	70.8	67.9	63.9	61.2	60.1	61.7	63.8	67.0	68.3	70.6	(27)
		MCDB	85.8	84.2	84.1	83.1	73.4	68.5	69.4	72.5	76.1	81.4	81.4	84.2	(28)
	2%	WB	70.2	70.2	68.8	66.0	62.2	59.3	58.4	59.4	61.3	64.6	66.0	68.7	(29)
		MCDB	82.1	81.5	80.3	77.4	69.7	64.5	65.7	66.4	70.9	76.4	77.4	80.5	(30)
	5%	WB	68.8	69.0	67.4	64.5	61.0	58.2	57.2	57.8	59.7	62.8	64.5	67.3	(31)
		MCDB	79.3	79.8	77.3	73.7	67.8	63.0	63.7	63.5	67.6	72.8	74.3	78.1	(32)
	10%	WB	67.4	67.7	66.0	63.2	60.0	57.1	55.9	56.5	58.2	61.2	63.1	66.1	(33)
		MCDB	76.8	77.4	74.7	70.8	65.8	61.8	61.5	61.7	64.8	69.5	71.6	75.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	17.6	18.1	18.8	17.4	16.9	17.8	16.8	16.9	18.0	16.8	16.7	(35)
		MCDBR	23.0	24.3	24.5	27.0	25.6	24.8	26.8	25.5	26.2	27.2	23.9	22.1	(36)
	5% WB	MCWBR	9.4	9.4	9.9	11.4	12.5	13.9	14.4	14.0	13.2	12.6	10.8	9.6	(37)
		MCWBR	19.1	20.0	20.3	21.8	19.1	17.5	19.8	19.6	21.7	21.8	19.6	19.0	(38)
(40) Clear-Sky Solar Irradiance	taub		0.341	0.343	0.334	0.322	0.320	0.314	0.316	0.332	0.341	0.351	0.338	0.342	(40)
		taud	2.538	2.539	2.560	2.591	2.589	2.576	2.565	2.518	2.454	2.443	2.523	2.525	(41)
	Ebn at Noon		316	309	300	287	270	262	268	278	292	302	315	317	(42)
		Edh at Noon	35	34	30	26	23	22	24	28	33	36	35	36	(43)
(44) All-Sky Solar Radiation	RadAvg		2598	2319	1861	1359	943	792	883	1121	1549	2041	2425	2625	(44)
	RadStd		60	68	72	67	59	46	52	54	64	96	73	65	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110	(46)
(47)	Regional (12 neighbors)	+0.65	+0.88	N/A	N/A	+0.95	+0.85	-4	-133	+200	+110	(47)

Nomenclature: See separate page